Suman Sourav
Jonathan J. Huerta y Munive

AALBORG UNIVERSITY
DENMARK

Models and Tools for Cyber-Physical Systems
Exercise sheet 7 - Scheduling
WITH SOLUTIONS

Exercise 1: Multi-processor cyclic scheduling

Consider a cyclic scheduling problem with the following tasks

Process	Period	Computation time
A	20	4
B	20	3
C	30	8
D	40	4
E	40	5
F	50	10
G	50	20

You can find the UPPAAL model of the scheduling task from the lecture in CyclicTasks.xml on Moodle.

(a) Adjust the model such that it represents this given scheduling problem. Then, use the model checking engine of UPPAAL to show that there does not exist any (cyclic) schedule for this problem.

(b) Adjust the model such that multiple task processors (CPUs) can perform the tasks in parallel. Ideally, create your model such that you can scale it easily to M processes.

(c) What is the minimum number of processors needed such that there exists a cyclic schedule for this problem?

.....Solution sketch .....

(a) After adjusting the model to this problem, the UPPAAL query

```
E[] not (PrA.ERR or PrB.ERR or PrC.ERR or PrD.ERR or PrE.ERR or
         PrF.ERR or PrG.ERR)
```

cannot be satisfied.

(b) A potential solution with a scalable number of processes can be found in CyclicTasksMultiProcess_solution.xml on Moodle.

(c) At least 2 processors are needed to obtain a cyclic schedule.

Exercise 2: Job Shop Scheduling

Jobshop scheduling problems involves a number of jobs $J_0, \ldots, J_n$ to be treated by a number of machines $M_0, \ldots, M_m$. In this particular variant each job J_i needs to be treated exactly once by each machine M_j and in a particular order and for a particular time-duration. The problem is to find an time-assignment (a schedule) indicating at which time J_i should be treated by M_j in a way such that the preferred sequence of J_i is respected and so that a machine M_j is never used by more than one job at a time. The challenge is to identify the minimum Makespan, i.e. time-assignment which the minimal total duration.

	Sport	Economy	Local News	Comic Stip
Kim	2. 5 min	4. 1 min	3. 3 min	1. 10 min
Jüri	1. 10 min	2. 20 min	3. 1 min	4. 1 min
Jan	4. 1 min	1. 13 min	3. 11 min	2. 11 min
Wang	1. 1 min	2. 1 min	3. 1 min	4. 1 min

Consider the particular Jobshop scheduling problem above, where 4 persons (Kim, Juri, Wang and Jan = the Jobs) needs to read 4 sections (Sport, Economy, Local News, Comic Strip = the Machines) of a single news paper. The timing in minutes as well as the preferred number in the reading sequence for each person is indicated in the cells.
You may use the UPPAAL model Jobshop.xml on Moodle of this instance of the jobshop.

(a) Use the model checking engine of UPPAAL to determine the minimum make-span, i.e. the minimum time until all persons has completed reading all sections.

(b) Identify also the minimum time for Kim to finish reading his sections, and the minimum time for Wang to finish his sections.

(c) What is the minimum time to have both Kim and Jan finishing reading their sections. Try to illustrate using informative Gantt charts.

.....Solution sketch .....

(a) The minimum make-span is 37. You can find this either by posing the UPPAAL query

```
E<> Kim.Done && Juri.Done && Jan.Done && Wang.Done
```

with diagnostic trace option "Fastest" activated (afterwards you find the time in the simulator). Alternatively you may search for the smallest value of `B` that will make the query `E<> Kim.Done && Juri.Done && Jan.Done && Wang.Done && time<=B` satisfied.

(b) Answer for `Kim` is 19 and for `Wang` is 4. Same technique to obtain this as above.

(c) Minimum time for both Kim and Jan to finish is 36 – so actually only 1 minute faster than the minimum time for having everyone finished reading the newspaper.